Akshit Kartik

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Education

The Pennsylvania State University, University Park, PA **Expected May 2026**
Bachelor of Science in Computer Science GPA: 3.56
Minor: Cybersecurity Computational Foundations
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Security, Databases, Web Development, Object-Oriented Programming

Publications

- **OmniSch: A Multimodal PCB Schematic Benchmark for Structured Diagram Visual Reasoning.** T. Lu, K. Lin, et al. (incl. A. Kartik). *arXiv preprint* arXiv:2604.00270 [cs.CV], 2026.
- **AquaTag: An Open-Source Wearable Platform Enabling Flexible Sensor Placement for Aquatic Sensing.** Co-author. *Proc. ACM IMWUT*, 2026. doi:10.1145/3831639.

Work Experience

Meritus.AI ***Chennai, India***
Research and Development Intern **June 2024 - August 2024**
- Developed and deployed YOLO-based object detection models onto the CV Pro AI Kit, utilizing OpenCV and Python to identify and track competition elements for the World Robot Olympiad (WRO) 2024.
- Engineered firmware in C++ for Arduino and ESP-32 microcontrollers to process sensor inputs, manage motor control, and execute real-time pathfinding logic for autonomous navigation.

The Pennsylvania State University ***University Park, PA***
Undergraduate Researcher **January 2025 - May 2026**
- Build the data pipeline for a CNN-LSTM gesture correction model, using OpenCV to preprocess video clips and Pandas to label and timestamp kinematic errors in sign language for downstream model training.
- Develop a custom Python annotation GUI for the AquaBit swimmer-tracking platform, synchronizing IMU sensor and video streams with NumPy/Pandas to produce a labeled multimodal dataset for activity recognition.

The Pennsylvania State University ***University Park, PA***
Learning Assistant **January 2024 - December 2025**
- Lead weekly recitation sections for 40+ students in a large-scale Java course, teaching object-oriented programming, data structures, and web application fundamentals.
- Provide targeted support by grading 100+ programming assignments, leading office hours, and delivering constructive feedback to improve code quality and design patterns.

Project Experience

J.A.R.V.I.S. - Agentic Voice & Text Personal Assistant **March 2026 - Present**
- Built a voice-and-text assistant on Claude's tool-use loop exposing 41 tools across 10 domains (Home Assistant, Spotify, TV, PC) behind an always-on WebSocket daemon, with episodic memory and barge-in streaming voice.
- Engineered a tiered, cost-aware model router (local Ollama → Claude Haiku → Sonnet) and gated every tool call through one audited security guard with permission tiers and turn-bounded confirmation that blocks prompt-injection self-approval; validated with 513 pytest tests.

AI-Powered Parts Procurement Engine **October 2025 - December 2025**
- Built a full-stack Next.js 15 / React 19 / TypeScript app for **B. Braun Medical Inc.** with NextAuth.js authentication, integrating the OpenAI API to automate sourcing of obsolete manufacturing components and replace a manual, weeks-to-months workflow.
- Designed a multi-field search and part-comparison engine that normalizes vendor specs and exports vetted matches to Excel via SheetJS; matched 100% of B. Braun's obsolete-part test catalog and cut per-part search time to 1–2 minutes.

OS Agent - Autonomous Windows System Optimizer **January 2026 - Present**
- Architected a Python agent that monitors CPU, RAM, GPU, and processes via psutil and triages anomalies through a LangGraph-orchestrated OODA state machine backed by a 4-layer decision engine and SQLite history.
- Integrated OpenAI GPT models with confidence-based escalation and a ChromaDB vector store of past fixes, eliminating redundant LLM calls on recurring anomalies under a configurable per-session cost cap.
- Hardened the agent with a fail-closed human-in-the-loop approval gate, command allowlisting/sanitization, protected-PID guards, and a Docker sandbox; verified with 163 pytest tests across decision, safety, and tooling layers.

Technical Skills

- **Languages:** Python, Java, C, C++, JavaScript, TypeScript, SQL, HTML/CSS, Bash
- **Frameworks & Libraries:** Next.js, React, NextAuth.js, LangGraph, PyTorch, TensorFlow/Keras, Scikit-learn, OpenCV, Pandas, NumPy
- **Tools & Platforms:** Git, GitHub, Docker, OpenAI API, ChromaDB, SQLite, REST APIs, Linux/Ubuntu, Windows